EarlyCare AI

Concept Note: An AI agent for early detection of chronic diseases

SDG 3: Good Health and Well-being | IBM SkillsBuild Masterclass 5 | October 2026

1. Project overview

EarlyCare AI is a free, multilingual AI agent that screens people for diabetes, heart disease, hypertension and chronic kidney disease in about 3 minutes and tells them what to do next. It combines a conversational symptom checker with machine learning models trained on basic health data, so risk is flagged before complications appear.

Item	Detail
Project title	EarlyCare AI: an AI agent for early disease detection
SDG	SDG 3: Good Health and Well-being (targets 3.4, 3.8, 3.c)
Project idea	AI agent for early disease detection using symptom checkers and ML on health data
Programme	IBM SkillsBuild, Masterclass 5: Creating real-life projects
Team / institution	[Add team names and college]
Deliverables	Lean Canvas (PDF), concept note, PowerPoint presentation

2. Problem statement

“Millions of Indians live with diabetes, heart disease, hypertension and kidney disease without knowing it, because screening is irregular and doctors are far away. This project builds an AI agent that uses a symptom checker and machine learning on basic health data to flag risk early and guide people to the right care.”

Chronic, non-communicable diseases (NCDs) are now India's biggest health burden, and most are found late:

- NCDs cause about two in three deaths. World Bank data puts NCDs at 65.9% of all deaths in India in 2019.
- A huge unscreened risk pool. The ICMR-INDIAB study (Lancet Diabetes & Endocrinology, 2023) estimates 101 million Indians with diabetes, 136 million with prediabetes and 315 million with hypertension.
- Specialists are out of reach. In March 2023 rural community health centres had 4,413 specialists against 21,964 required, a shortfall of about 80% (Health Dynamics of India 2022-23).

Why detection is late: symptoms are silent in early stages; tests mean travel and lost wages; most health apps are English-only; and one-time screening camps rarely follow up.

Existing alternatives and their gaps:

Alternative	Gap
Google search / global symptom-checker apps	English-first, no Indian context, no follow-up
Paid full-body check-up packages	Costly, city-based, done rarely
Government NCD screening camps	Infrequent, limited reach, paper records
Waiting for symptoms	Late diagnosis and expensive treatment

3. Objectives

Build a conversational AI agent that collects symptoms, lifestyle and family history in Kannada, Hindi and English, by chat or voice.

Train and validate ML models that estimate risk for type 2 diabetes, heart disease, chronic kidney disease and hypertension, with recall of 85% or more.

Explain every risk score in simple words, showing the main factors behind it.

Guide each user to a clear next step: nearest PHC or lab, teleconsultation, or healthy-habit advice, with follow-up reminders.

Equip ASHA workers to screen whole households and share summaries with their PHC.

Pilot with 5,000 people in one Karnataka district within six months.

4. Proposed solution

EarlyCare AI works like a health-check camp in your pocket: it talks to people in their own language, scores their risk, and points them to care. It screens and guides; it never diagnoses.

How it works (user journey)

Step	What happens
1. Start	The user opens the WhatsApp bot or Android app and picks a language.
2. Answer	The agent asks about symptoms, lifestyle, diet, activity and family history.
3. Measure (optional)	BP, blood sugar and weight from a health camp or home device are added.
4. Predict	ML models return a low, medium or high risk level for each condition, with the reasons.
5. Next step	The agent suggests a confirmatory test, a doctor visit or healthy habits, sets reminders, and checks in again in 3 to 6 months.

Key features

- Local-language chat and voice for low-literacy users
- Risk scoring for four chronic diseases from simple inputs
- Explainable results (for example, “high BMI and family history raised your diabetes risk”)
- Care finder for the nearest PHC, government lab or teleconsultation
- Follow-up reminders for tests, medicines and re-screening
- ASHA worker mode to screen a household and share a summary with the PHC

Unique value proposition: know your risk before it becomes a disease, in your own language, in about 3 minutes, for free.

5. Technical approach

The system has five parts, and a request flows through them in order:

- User channels: WhatsApp bot, Android app, and a web app for ASHA workers and PHCs.
- AI conversation agent: an IBM Granite model on IBM watsonx.ai runs the dialogue; Bhashini APIs handle Indian-language translation and speech.
- ML risk engine: four trained models score risk, and SHAP explains each prediction.
- Care and reminders: PHC and lab locator, teleconsultation links and scheduled nudges.
- Secure data store: encrypted, consent-based records in IBM Cloudant.

The backend is a Python FastAPI service on IBM Cloud Code Engine, with HTTPS and role-based access between services.

Data and models

Condition	Training data (public)	Key inputs	Models
Type 2 diabetes	PIMA Indians Diabetes dataset; Indian Diabetes Risk Score (MDRF)	Age, BMI, waist, glucose, family history	Logistic Regression, Random Forest

Condition	Training data (public)	Key inputs	Models
Heart disease	UCI Heart Disease (Cleveland)	Age, BP, cholesterol, chest pain, smoking	Random Forest, XGBoost
Chronic kidney disease	UCI Chronic Kidney Disease dataset	BP, sugar, albumin, swelling, diabetes	Random Forest
Hypertension	Clinical BP thresholds + risk questionnaire	BP readings, salt intake, stress, age	Rule-based + Logistic Regression

Evaluation: recall (sensitivity) is the main metric, so few at-risk people are missed; ROC-AUC and precision are also tracked, using k-fold cross-validation.

Technology stack: Python, pandas, scikit-learn, XGBoost, SHAP; IBM watsonx.ai and Granite models; Bhashini APIs; FastAPI on IBM Cloud Code Engine; WhatsApp Business API, Android, Streamlit dashboard; IBM Cloudant.

6. Target users

- Individuals: adults 30+ in Tier-2/3 towns and villages, especially with a family history of diabetes or BP.
- Frontline workers: ASHA and ANM workers who screen communities.
- Institutions: PHCs, state health missions and NGOs.
- Payers: employers, insurers and diagnostic labs.

Early adopters: smartphone users aged 30-60 who already worry about sugar or BP; ASHA workers in one pilot district in Karnataka; NGOs and college NSS units running free health camps.

7. Business model

Core screening is always free for individuals. Revenue comes from those who benefit at scale:

Stream	How it works
B2G licences	Per-screening fee or annual licence for state health missions and PHCs
B2B wellness plans	Per-employee pricing for employers and insurers
Lab referrals	Fee from partner labs for confirmatory tests booked through the app
CSR and grants	Health-tech CSR funds and innovation grants for rural roll-out

Main costs: cloud hosting and AI inference, app and model development, clinical review, data security and compliance, ASHA training and outreach.

8. Implementation plan

Phase	Months	Activities
Research and data	1-2	Study datasets, consult a doctor, design questions in 3 languages
Build MVP	3-4	Train and validate models, build the agent on watsonx.ai, launch WhatsApp bot
Pilot	5-6	Test with ASHA workers and one NGO camp in a Karnataka district
Improve and scale	7-12	Add languages and diseases, PHC dashboard, partner onboarding

9. Expected impact and key metrics

- 5,000 people screened in the six-month pilot, rising to 50,000 in year 1

- Model recall of 85% or more on validation data
- 50% of high-risk users take a confirmatory test within 30 days
- Cost per screening under Rs 10
- Earlier diagnosis, lower treatment costs and progress on SDG target 3.4

10. Ethics, privacy and safety

- Not a diagnosis: clear disclaimers; high-risk results always route to a doctor or PHC.
- Privacy by design: consent first, minimal data, encryption, aligned with India's Digital Personal Data Protection Act, 2023.
- Fairness: model accuracy checked across age, gender and rural/urban groups.
- Human oversight: a clinical advisor reviews questions, thresholds and red-flag rules.
- Inclusion: voice and local languages for low-literacy and elderly users.

11. Conclusion

EarlyCare AI turns a phone into an early-warning system for chronic disease. By combining a local-language AI agent with explainable ML, it helps people find risk early, get tested and stay on track, supporting SDG 3 where specialists are scarce.

Sources

- ICMR-INDIAB study, The Lancet Diabetes & Endocrinology (2023): 101 M diabetes, 136 M prediabetes, 315 M hypertension.
- World Bank Development Indicators: cause of death by non-communicable diseases, India, 2019 (65.9%).
- Ministry of Health and Family Welfare, Health Dynamics of India (Infrastructure & Human Resources) 2022-23.